AB³ACBS 2016

A festival of bioinformatics

COMBINE

COMBINE STUDENT SYMPOSIUM

Monday, 31 October

Time	Activity
8.30-9.30am	COMBINE registration and poster hanging
9.00-9.55am	Welcome and opening remarks
	• **Soroor Hediyeh-Zadeh** Computational workflows for research students: towards a reproducible research • **Nesli Avgan** Applying machine learning to GWAS analysis of major memory traits
9:55-10:25am	Poster Lightning Talks 1
10:25 – 12:00pm	Morning tea and poster viewing
	• **Momeneh Foroutan** A transcriptional signature for TGFβ-induced epithelial-mesenchymal transition in cancer • **Luke Zappia** Simplifying simulation of single-cell RNA sequencing • **Ramyar Molania** Accurate and robust normalization of Nanostring nCounter gene expression data
12:00-1:15pm	Lunch and poster viewing
1:15-1:30pm	Group photo
	• **Andrew Pattison** Investigating the 3' untranslated region of mRNA in order to understand the drivers of metastasis in primary triple negative breast tumours • **Areej Alsheikh-Hussain** Insertion sequence elements are drivers of diversification in the broad host range aquatic pathogen Streptococcus iniae • **Son Hoang Nguyen** Scaffolding and completing genome assemblies in real-time with nanopore sequencing
2:30-3:00pm	Poster Lightning Talks 2
3:00-3:30pm	Afternoon tea at Botanic Bar
3:30-5:00pm	• Panel discussion **David Lovell, Katherine Pillman, Lynn Fink, Nicola Waddell and CX Chan**
5:00-5:15pm	Talk and poster prizes, Symposium close
5:15-6:30pm	Free time
6:30pm- late	COMBINE Trivia night and social event - Botanic Bar

AB³ACBS CONFERENCE, DAY 1

Tuesday, 1 November

Time	Activity
8:00-9:00am	AB3ACBS registration and poster hanging
9:00-9:15am	Housekeeping; Welcome and Opening by **Queensland Chief Scientist – Dr Geoff Garrett**
9:00-10:30am	**SESSION 1:** Microbes
	• **Nouri Ben Zakour** Evolutionary epidemiology of successful bacterial pathogens • **Andrew Buultjens** Exploiting extremes of Legionella pneumophila genomic diversity for accurate source attribution • **Darryl Reeves** Probabilistic Inference and Shared Parameter Learning for Metagenomic Sequence Analysis • **Patrick Laffy** HoloVir: Taxonomic and functional analysis of viral metagenomic communities.
10:30-11:00am	Morning tea
11:00-12:30pm	**SESSION 2:** Cancer
	• **Andreas Schreiber** Taking the confusion out of fusions: Structural mutation detection in cancer research and diagnostics • **Shila Ghazanfar** Exploring Pan-Cancer Network Relationships Between Somatic Changes and Expression Profiles with PACMEN • **Luis Lara-Gonzalez** Consequences of Drug Dose Modulation on Clonal Dynamics • **Rebecca C. Poulos** Functional mutations form at CTCF/cohesin binding sites in melanoma due to uneven nucleotide excision repair across the motif • **Christoffer Flensburg** Tracking clonal evolution in cancer from multiple samples
12:30-1:30pm	Lunch
1:30-3:00pm	**SESSION 3:** Statistics
	• **Matt Ritchie** Tools for comparing and combining RNA-seq results • **Paul Lin** CIDR: Ultrafast and accurate clustering through imputation for single-cell RNA-Seq data • **Belinda Phipson** Gene length bias in single cell RNA-seq data • **Joseph Cursons** Computational methods to examine micro-RNA targeting of interacting protein networks • **Göknur Giner** FRY: A Fast Approximation to ROAST Gene Set Test with Mean Aggregated Set Statistics
3:00-3:30pm	Afternoon tea
3:30-5:00pm	**SESSION 4:** Plants and Animals
	• **Kate Hertweck** Plant systematics to cancer biology: genome-wide patterns and organismal evolution • **Jimmy Breen** Whole methylome analysis of grapevine reveals tissue specific DNA methylation variation • **Andrew Lonsdale** Mighty Morphin FASTA Files • **Nadia Davidson** SuperTranscript: a compact reference for the transcriptome • **Stephen Fletcher** Small Complementary RnA Mapper (SCRAM): a tool for studying small interfering RNA biogenesis in plants
5:00pm	**Poster Session**
6:30pm	**Conference Dinner**

AB^{3}ACBS CONFERENCE, DAY 2

Wednesday, 2 November

Time	Activity
9:00-10:30am	**SESSION 5**: Evolution
	• **Simon Ho** Phylogenomic analysis and molecular evolutionary clocks • **Åsa Pérez-Bercoff** Investigating the evolution of new biochemical pathways in baker's yeast Saccharomyces cerevisiae • **Miles Benton** Identification of allelic-specific methylation profiles across generations in the Norfolk Island genetic isolate • **Kevin Murray** Estimating genetic similarity with the k-mer Weighted Inner Product (kWIP) • **Martin Smith** De novo characterisation of RNA structure motifs from ENCODE RIPseq data
10:30-11:00am	Morning tea
11:00-12:30pm	**SESSION 6**: 'Omics
	• **Ute Roessner** Metabolomics - an important piece in the 'omics puzzle • **Pip Griffin** EMBL Australia Bioinformatics Resource (EMBL-ABR) • **Jovana Maksimovic** Through the looking glass: using Monocle to visualise methylation array data • **Ruth Fuhrman-Luck** Big data from a little proteolysis: Combining multiple omics platforms to identify novel functions of a protease associated with prostate cancer • **Lawrence Buckingham** Geometric map-reduce algorithms for alignment-free sequence comparison
12:30-1:00pm	Lunch
1:00-2:00pm	ABACBS Annual General Meeting
2:00-3:40pm	**SESSION 7**: Disease
	• TBA COMBINE Student • **Daniel Cameron** GRIDSS: sensitive and specific genomic rearrangement detection using positional de Bruijn graph assembly • **Harriet Dashnow** Detecting pathogenic STR expansions in next-gen sequencing data • **Rod Lea** BootNet: a bootstrapping application for GLMnet modelling to identify robust classifiers in genomics data • **Alexis Lucattini** Assessing the Practicality of Oxford Nanopore Sequencing in Clinical Diagnositcs
3:40-5:00pm	**SESSION 8**: Systems
	• **Anna Trigos** How do vulnerabilities left during the evolution of cellular networks set the stage for cancer? • **Stuart Archer** TCP-seq, a novel technique for investigating mechanisms and regulation of eukaryotic translation initiation • **Teresa Attwood** The Road to Utopia: challenges in linking literature & research data
5:00pm	**Close**

Connecting to the QUT Conference Wireless Network

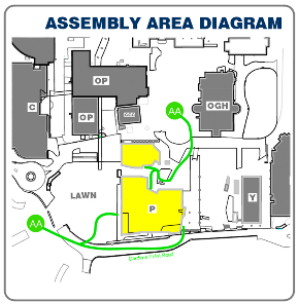
If you have an eduroam account use the '**eduroam**' wireless network as you would at your institution.

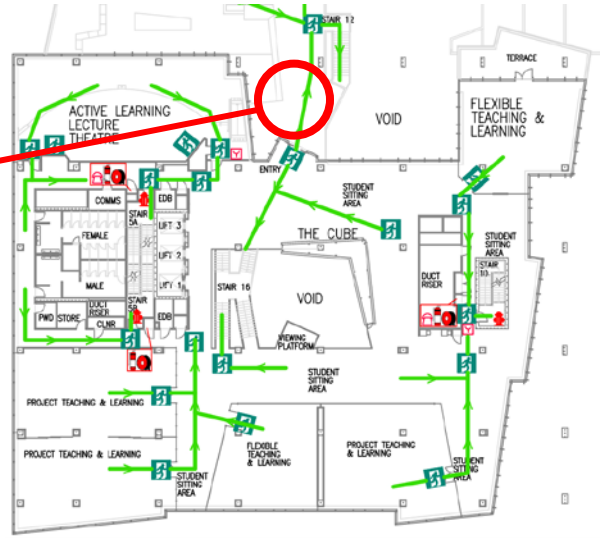
Via a mobile device (phone/tablet)	Via a laptop (PC / Mac)
1. Go to your device's '**Settings**' menu 2. Select '**Wi-Fi**' 3. Select the network called '**Events@QUT**' note – If you have used this service before you will have to "Forget this Network". 4. When prompted for a password, enter: **Myz9012M** (note – password is case-sensitive) 5. Open any web browser to http://www.qut.edu.au/ (note – do not use https://) and a page titled '**Welcome to Events@QUT**' will appear, displaying the service Terms and Conditions. 6. Select the '**Click to Accept**' the Terms and Conditions option to commence using the conference wireless service.	1. Go to your wireless network connections, by clicking on the '**wireless**' logo on your laptop. 2. Select the wireless network called '**Events@QUT**' note – If you have used this service before you will have to "Forget this Network". 3. When prompted for a password, enter: **Myz9012M** (note – password is case-sensitive) 4. Open any web browser to http://www.qut.edu.au/ and a page titled '**Welcome to Events@QUT**' will appear, displaying the service Terms and Conditions. 5. Select the '**Click to Accept**' the Terms and Conditions option to commence using the conference wireless service.

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